

MIAE 221: Materials Science for Engineers

Master Lecture Notes Fill-in-the-Blank Solutions Key (Lectures 1–7)

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Executive Overview

Dr. Medraj's lecture slide decks for MIAE 221 are intentionally prepared with dotted underlines (.....), blank entries, and prompt questions designed for students to complete during lecture.

This master guide reviews **every single blank, missing word, numerical constant, and incomplete table across Lectures 1 through 7**, filling them in with 100% precision from the course textbook (

Callister: Materials Science and Engineering: An Introduction

) and engineering science principles.

Quick Reference: Your Specific Question

Slide Prompt (Lecture 7, Slide 13 — Hume-Rothery Rule 1):

"For appreciable substitutional solubility to occur the following factors must hold:

- *Atomic size factor: the difference in atomic radii between the two atoms must be $\leq$ "*

The Answer: $\leq \pm 15$ (or simply 15)

- **Textbook Reference:**

Callister (10th Ed.)

, Section 4.3 (page 112):

$$\Delta R = \left| \frac{R_{\text{solute}} - R_{\text{solvent}}}{R_{\text{solvent}}} \right| \times 100$$

- **Physical Reason:** If the solute atom's radius differs by more than 15, severe local lattice strain (compressive if oversized, tensile if undersized) destabilizes the host lattice, causing the system to separate into a distinct second phase rather than forming a substitutional solid solution.
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Lecture 1: Introduction to Materials Science & Engineering

Slide #	Professor's Incomplete Prompt / Blank Line	Filled-in Text / Answer (Callister & Lecture Context)	Core Scientific Concept & Exam Significance
Slide 4	<i>Oils, gases, pharmaceuticals:</i> <i>Iron, copper, polymers, cement:</i>	Non-materials (chemicals, fluids, foodstuffs) Materials (engineering structural/functional solids)	Materials are solids with specific geometric form and engineering load-bearing or functional utility.
Slide 7	<i>What is Materials Science?</i> Relationships between structure and of materials. <i>What is Materials Engineering?</i> Designing the structure to impart properties.	properties predetermined / desired	Science investigates basic structure ↔ property causality; Engineering applies that causality to design products.
Slide 8	<i>Property:</i> Response of a material to an effect.	external stimulus / external effect	E.g., mechanical load → deformation; voltage → current; heat → expansion. Properties are independent of size and shape.

Lecture 2: Review of Chemistry & Bonding Foundations

Slide #	Professor's Incomplete Prompt / Blank Line	Filled-in Text / Answer	Core Scientific Concept & Textbook Reference
Slide 2	<i>Materials Property Comparison Table:</i> • Conductivity (Electrical & Thermal) for Polymers: • Corrosion Resistance for Ceramics: • Machinability for Ceramics:	• Polymers: Poor / Very Poor (insulators) • Ceramics: Exceptional / Good (inert to chemical attack) • Ceramics: Poor / Low (extreme hardness & brittleness)	Callister Table 1.1: Ceramics have superior chemical and thermal resistance but are notoriously difficult to machine due to brittleness.
Slide 3	• Number of electrons and (tendency for an atom to attract an electron). • Also by how atoms are arranged in space →	electronegativity crystal structure (crystallography)	Electronegativity dictates primary bond type; 3D spatial packing dictates crystal lattice and macroscopic anisotropy.
Slide 12	• Avogadro's Number $N_A =$ part/mole	6.022×10^{23} atoms/mole	Fundamental conversion between atomic mass units (amu) and macroscopic grams (g).
Slide 14	• <i>Most elements:</i> • Electron configuration • <i>Stable configurations tend to be:</i> •	incomplete / unclosed outer subshells inert / chemically unreactive (noble gases)	Full <i>s</i> and <i>p</i> octets (s^2p^6) have zero chemical driving force to bond (He, Ne, Ar, Kr, Xe).
Slide 16	• Deep potential well → bonded. • Shallow well → bonded.	strongly bonded (high E_0 , high T_m) weakly bonded (low E_0 , low T_m)	Callister Section 2.5: Deep wells require substantial thermal kinetic energy to overcome atomic bonding.
Slide 17	<i>State as function of bonding energy:</i> • Solid: (.....) • Liquid: (.....) • Gaseous: (.....) <i>The higher the bond energy:</i> 	• Solid: large / strong bonding energy ($E_0 > 100$ kJ/mol) • Liquid: moderate / intermediate bond energy • Gaseous: weak / low bond energy ($E_0 < 10$ kJ/mol) • Higher melting point (T_m), higher modulus (E), lower expansion (α)	Fundamental interatomic well properties: deeper well → higher T_m ; steeper curvature → higher stiffness E ; symmetric curve → low α .

Lecture 3: Chemical Bonding Types & Physical Properties

Slide #	Professor's Incomplete Prompt / Blank Line	Filled-in Text / Answer	Core Scientific Concept & Textbook Reference
Slide 2	<i>Ionic Bonding:</i> Large difference in is required.	electronegativity ($\Delta X > 1.7$)	Metal easily surrenders valence electrons to non-metal, forming electrostatic +/– Coulombic attraction.
Slide 3	Predominant bonding in:	ceramics (compounds of metal + nonmetal)	Callister Section 2.6: Oxides, nitrides, carbides are predominantly ionic.
Slide 4	<i>Typical characteristics of ionically-bonded materials:</i> melting temperature (electrical and thermal conductivity)	High melting temperature Insulators / Low conductivity	Localized electrons cannot drift freely; high electrostatic force requires immense thermal energy to dissociate.
Slide 5	<i>Covalent bonding:</i> Requires electrons.	sharing of (directional covalent bonds)	Carbon has 4 valence electrons; shares 4 electrons to complete stable sp^3 octet (diamond, polymers).
Slide 10	<i>Metallic Bonding:</i> Valence electrons are not bound to any specific atom but are to drift.	free to drift (forming an electron sea / cloud)	Non-directional metallic bond enables atomic planes to slide easily, imparting high ductility and high conductivity.
Slide 13	- The hydrogen bond is generally stronger than - Polar molecule induced dipoles & bonding.	van der Waals bonds (dispersion / London forces) hydrogen bonding	Hydrogen bonding is a strong permanent dipole interaction between H and highly electronegative O, N, F.

Slide #	Professor's Incomplete Prompt / Blank Line	Filled-in Text / Answer	Core Scientific Concept & Textbook Reference
Slide 18	<i>Bonding Type Classification for Materials:</i> • Brass: • Rubber: with some • BaS: predominantly • Solid Xenon: • Nylon: with some • AIP: predominantly	• Brass: metallic (Cu-Zn alloy) • Rubber: covalent (chain backbone) with van der Waals (between chains) • BaS: ionic ($\Delta X = 2.58 - 0.89 = 1.69, \sim 50$ ionic) • Solid Xenon: van der Waals / induced dipole (inert gas) • Nylon: covalent (backbone) with hydrogen bonding (amide groups) • AIP: covalent ($\Delta X = 2.19 - 1.61 = 0.58$, III-V semiconductor)	Callister Problem 2.18 & Lecture 3 summary table: Master exam classification of materials bonding.

Lecture 4: Crystal Structures 1 (Unit Cells, APF, CN)

Slide #	Professor's Incomplete Prompt / Blank Line	Filled-in Text / Answer	Core Scientific Concept & Textbook Reference
Slide 2	- Non-dense, packing. - Dense, packing.	random / amorphous packing regular / crystalline packing	Dense, ordered crystalline structures minimize interatomic energy, creating stable solid lattices.
Slide 3	 = Noncrystalline.	Amorphous (e.g., non-crystalline vitreous SiO_2 glass)	Lacks long-range translational atomic periodicity.
Slide 6	- Hard sphere model of a crystal system. - The diagram represents a simple cubic	Simple Cubic (SC) unit cell	Atoms touch only along cube edges: $a = 2R$.
Slide 7	- A unit cell is the smallest entity that exhibits the chemical and properties of the material. - The length of each unit cell axis is called a	physical (crystallographic) lattice parameter (a, b, c)	Repeating unit cells in 3D generate the entire macroscopic crystal lattice.
Slide 8	Simple cubic structure contains ... atom/unit cell.	1 atom/cell ($8 \text{ corners} \times \frac{1}{8} = 1$)	Only corner atoms; face and center are vacant.
Slide 9	- Adding an atom at center gives a (BCC) structure. - Atoms touch along the body - BCC unit cells have a CN =	Body-Centered Cubic diagonal ($\langle 111 \rangle$) CN = 8	Coordination number CN is the number of nearest touching neighbors ($a\sqrt{3} = 4R$).
Slide 10	The packing efficiency (APF) of a BCC structure is	68 (APF = 0.68)	$\text{APF} = \frac{2 \times \frac{4}{3}\pi R^3}{(4R/\sqrt{3})^3} = \frac{\pi\sqrt{3}}{8} \approx 0.680$.
Slide 11	Atoms at corners and face centers: (FCC) crystal structure.	Face-Centered Cubic	Touching along face diagonals ($\langle 110 \rangle$, $a\sqrt{2} = 4R$).
Slide 12	- Number of atoms in FCC: - FCC unit cells have CN = and APF =	4 atoms/cell ($8 \times \frac{1}{8} + 6 \times \frac{1}{2} = 4$) CN = 12 and APF = 0.74 (74)	Maximum possible packing efficiency for equal-sized rigid spheres.
Slide 13	- HCP represents a structure. - Number of atoms in HCP = - HCP Coordination Number CN =	Hexagonal Close-Packed 6 atoms/cell ($12 \times \frac{1}{6} + 2 \times \frac{1}{2} + 3 = 6$) CN = 12 (and APF = 0.74)	Ideal $c/a = \sqrt{8/3} \approx 1.633$.

Lecture 5: Crystal Structures 2 (Miller Indices & Crystallography)

Slide #	Professor's Incomplete Prompt / Blank Line	Filled-in Text / Answer	Core Scientific Concept & Textbook Reference
Slide 2	<i>Simple Cubic Review:</i> • CN = • Close-packed directions: • Number of atoms: • APF =	• CN = 6 • Close-packed: cube edges / $\langle 100 \rangle$ • Atoms/cell: 1 atom • APF = 0.52 (52)	Low packing factor makes Simple Cubic unstable for pure metals, with the sole exception of α -Polonium.
Slide 3	<i>BCC Review:</i> • Unit cell: • CN = • Close-packed directions: • APF =	• Unit cell: BCC • CN = 8 • Close-packed: body diagonals / $\langle 111 \rangle$ • APF = 0.68 (68)	Examples: α -Fe, Cr, W, Mo, Ta.
Slide 4	<i>FCC Review:</i> • Unit cell: • CN = • Close-packed directions: • APF =	• Unit cell: FCC • CN = 12 • Close-packed: face diagonals / $\langle 110 \rangle$ • APF = 0.74 (74)	Examples: Cu, Al, Ni, Au, Ag, γ -Fe.
Slide 6	<i>HCP Review:</i> • Unit cell: • CN = • APF = • Stacking Sequence:	• Unit cell: HCP • CN = 12 • APF = 0.74 • Stacking: ABABAB...	In contrast, FCC has stacking sequence ABCABC...
Slide 9	• Deformation under loading (.....) occurs on certain crystalline planes and directions. • Directions and planes are described using three integers: Indices	slip / plastic deformation Miller Indices ($[uvw]$ for directions, (hkl) for planes)	Dislocations move preferentially along close-packed directions on close-packed planes.

Lecture 6: Crystal Structures 3 (Densities, Anisotropy, XRD, Allotropy)

Slide #	Professor's Incomplete Prompt / Blank Line	Filled-in Text / Answer	Core Scientific Concept & Textbook Reference
Slide 4	FCC [100] Linear Density: - For FCC $a = \dots\dots\dots$ $LD = \dots\dots\dots$	$a = 2\sqrt{2}R$ $LD_{[100]} = \frac{1}{a} = \frac{1}{2\sqrt{2}R}$	Vector passes through 2 corner atoms: $n = 2 \times \frac{1}{2} = 1$ atom. Length $L = a = 2\sqrt{2}R$.
Slide 6	FCC (110) Planar Density: - Number of atoms = $\dots\dots$ $PD = \dots\dots\dots$	- Number of atoms = 2 atoms ($4 \times \frac{1}{4} + 2 \times \frac{1}{2} = 2$) $PD_{(110)} = \frac{1}{4\sqrt{2}R^2}$	Plane area is rectangle: $A = a \times a\sqrt{2} = \sqrt{2}a^2 = 8\sqrt{2}R^2$.
Slide 7	- Slip ($\dots\dots\dots$ in metals) depends on linear and planar density. - Slip occurs on planes that have the $\dots\dots\dots$ density.	plastic deformation greatest / highest atomic planar density	Close-packed planes have the largest interplanar spacing d_{hkl} , minimizing the critical resolved shear stress required for slip.
Slide 8	Most engineering materials are $\dots\dots\dots$	polycrystalline (aggregates of many crystalline grains)	Grain boundaries impede dislocation motion, strengthening the metal (Hall-Petch effect).
Slide 9	- Single Crystals: Properties vary with direction: $\dots\dots\dots$ - If grains are randomly oriented: $\dots\dots\dots$	anisotropic (directional dependence) macroscopically isotropic	E.g., BCC Iron: $E_{[111]} = 272.7$ GPa, $E_{[100]} = 125.0$ GPa; random polycrystal averages to $E = 210$ GPa.
Slide 11	Have a spacing comparable to the $\dots\dots\dots$ of the incident wave.	wavelength (λ) (~ 0.1 nm or $1 \text{ \AA}$)	Diffraction occurs only when obstacle spacing is on the order of radiation wavelength.
Slide 13	Constructive & Destructive Interference: In-phase waves $\dots\dots\dots$; out-of-phase waves $\dots\dots\dots$	reinforce (constructive, high peak intensity) cancel (destructive, zero intensity)	Bragg condition: $n\lambda = 2d \sin \theta$.
Slide 17	Aluminum FCC (110) peak evaluation: $\theta = \dots\dots\dots^\circ$ $2\theta = \dots\dots\dots^\circ$	$\theta = 19.3^\circ$ $2\theta = 38.5^\circ$	Diffractometers always measure 2θ (deviation angle from incident beam).

Lecture 7: Imperfections in Solids (Defects, Solid Solutions, Dislocations)

Slide #	Professor's Incomplete Prompt / Blank Line	Filled-in Text / Answer	Core Scientific Concept & Textbook Reference
Slide 2	• “ ” lower the free energy of a crystal & make it more stable. • Defect does not necessarily imply a thing.	Defects / Imperfections (increase entropy: $\Delta G = \Delta H - T\Delta S$) bad / detrimental thing	E.g., carbon in iron forms steel; impurities in silicon create transistors and microprocessors.
Slide 4	• Self-interstitials are far less common than vacancies because of the relatively energy required.	high activation / high strain energy	Forcing a large host atom into a tiny void severely strains the surrounding lattice, making formation energy $Q_i \gg Q_v$.
Slide 7	• $\exp(-Q/kT)$ is an function of temperature.	exponential / Arrhenius	Vacancy concentration increases exponentially with temperature: $N_v = N \exp(-Q_v/k_B T)$.
Slide 9	• Add 1 at. to Pb → Result: • Add 25 at. to Pb → Result:	substitutional solid solution (single phase) two-phase / multiphase alloy (Sn solubility limit exceeded)	At 25 at., the solubility limit is exceeded, precipitating separate Sn-rich second-phase particles.
Slide 10	• Sn islands move along surface to alloy Cu to make • Islands move into regions.	bronze (Cu-Sn alloy) unalloyed / pure Cu	Solid-state surface diffusion driving spontaneous alloying.
Slide 12	<i>Second phase particles have:</i> • different • often different	chemical composition crystal structure (and lattice parameter)	E.g., Fe_3C (cementite, orthorhombic) in $\alpha\text{-Fe}$ (ferrite, BCC).
Slide 13 (From User Image)	Hume-Rothery Rules for Substitutional Solid Solutions: 1. <i>Atomic size factor:</i> difference in radii must be $\leq$ 2. <i>Electronegativity:</i> difference must be $\leq$ 3. <i>Crystal structure:</i> must be the 4. <i>Valence:</i>	1. $\leq \pm 15$ (or 15) 2. $\leq \pm 0.4$ eV (large $\Delta X \rightarrow$ intermetallics) 3. same (e.g., both FCC for complete Cu-Ni solubility) 4. higher valency dissolves more readily in lower valency	The 4 Cardinal Hume-Rothery Rules. Complete mutual solubility occurs only when all four rules are simultaneously satisfied!

Slide #	Professor's Incomplete Prompt / Blank Line	Filled-in Text / Answer	Core Scientific Concept & Textbook Reference
	solute with valency is more soluble.		
Slide 19	• Dislocations having both edge and screw character: dislocations.	mixed	In real alloys, dislocation lines are curved loops: pure edge where $\vec{b} \perp \vec{t}$, pure screw where $\vec{b} \parallel \vec{t}$, and mixed in between.

Pedagogical Summary & Midterm Advice

- Calculations:** Always memorize the $a \leftrightarrow R$ relations:
 - BCC:** $a\sqrt{3} = 4R \implies a = \frac{4R}{\sqrt{3}}$
 - FCC:** $a\sqrt{2} = 4R \implies a = 2\sqrt{2}R$
- Hume-Rothery Rules:** If an exam question asks whether an alloy forms a solid solution, test all 4 rules:
 - Cu-Ni:** Radius difference 2.3, both FCC, $\Delta X = 0.1$, compatible valency $\implies$ **Complete isomorphous solid solution.**
 - Cu-Zn:** Crystal structures differ (FCC vs HCP) $\implies$ **Limited solubility only** (≤ 35 wt).
- Vacancy Activation Energy:** Plotting $\ln(N_v/N)$ versus $1/T$ yields a straight line with slope $= -Q_v/k_B$.
Use $k_B = 8.617 \times 10^{-5}$ eV/K when Q_v is given in eV.